

Solutions to Exercise Sheet 1: Groups and Lie Algebras

Exercise 1: The symmetric group S_n

Recall that S_n is the group of permutations of the set $\{1, 2, \dots, n\}$. We write a permutation σ as

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

A cycle $(a_1, a_2, \dots, a_k)$ sends $a_1 \mapsto a_2$, $a_2 \mapsto a_3$, $\dots$, $a_k \mapsto a_1$, and fixes all other elements.

1(a) Decomposition into disjoint cycles

Let $\sigma \in S_n$. We show that σ can be written as a product of disjoint cycles.

Pick any element $i \in \{1, 2, \dots, n\}$ and consider its successive images under σ :

$$i, \sigma(i), \sigma^2(i), \sigma^3(i), \dots$$

Since the set $\{1, 2, \dots, n\}$ is finite, this sequence cannot consist of distinct elements forever. Hence there exist integers $r < s$ such that

$$\sigma^r(i) = \sigma^s(i).$$

Because σ is a permutation, it is invertible. Applying σ^{-r} to both sides gives

$$i = \sigma^{s-r}(i).$$

Therefore, starting from i and repeatedly applying σ , we eventually return to i . Let $k \geq 1$ be the smallest positive integer such that

$$\sigma^k(i) = i.$$

Then the elements

$$i, \sigma(i), \sigma^2(i), \dots, \sigma^{k-1}(i)$$

are pairwise distinct, and they form the cycle

$$(i, \sigma(i), \sigma^2(i), \dots, \sigma^{k-1}(i)).$$

Now choose an element not yet contained in this cycle and repeat the same construction. Since the set is finite, this process ends after finitely many steps, and every element belongs to exactly one such cycle. Thus σ is a product of cycles.

These cycles are disjoint: if two cycles had an element in common, then starting from that element and repeatedly applying σ would generate the same orbit, so the two cycles would in fact be the same.

Hence every permutation can be written as a product of disjoint cycles.

Example

Let

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix}.$$

We compute orbits:

$$1 \mapsto 3 \mapsto 5 \mapsto 6 \mapsto 2 \mapsto 1$$

Thus:

$$(1, 3, 5, 6, 2)$$

Remaining element:

$$4 \mapsto 4 \Rightarrow (4)$$

Hence:

$$\sigma = (1, 3, 5, 6, 2)(4)$$

1(b) Every cycle is a product of transpositions

A transposition is a cycle of length 2. Let

$$(a_1, a_2, \dots, a_k)$$

be a cycle of length k . We claim that

$$(a_1, a_2, \dots, a_k) = (a_1, a_k)(a_1, a_{k-1}) \cdots (a_1, a_3)(a_1, a_2).$$

We verify this by checking the action on the elements $a_1, a_2, \dots, a_k$.

First, a_1 is sent by the rightmost transposition (a_1, a_2) to a_2 , and no other transposition on the left moves a_2 . Hence overall

$$a_1 \mapsto a_2.$$

Next, a_2 is sent by (a_1, a_2) to a_1 , then by (a_1, a_3) to a_3 , and then no later transposition moves a_3 . Hence

$$a_2 \mapsto a_3.$$

Similarly,

$$a_3 \mapsto a_4, \quad \dots, \quad a_{k-1} \mapsto a_k.$$

Finally, a_k is fixed by all transpositions except the leftmost one (a_1, a_k) , which sends it to a_1 . Thus

$$a_k \mapsto a_1.$$

All elements not among $a_1, \dots, a_k$ are fixed by every transposition on the right-hand side.

Therefore the right-hand side acts exactly as the cycle $(a_1, a_2, \dots, a_k)$.

Since every permutation is a product of disjoint cycles by part (a), and every cycle is a product of transpositions, every permutation is a product of transpositions. Hence S_n is generated by transpositions.

Example

$$(1, 3, 5, 6, 2) = (1, 2)(1, 6)(1, 5)(1, 3)$$

Check:

$$1 \mapsto 3, \quad 3 \mapsto 5, \quad 5 \mapsto 6, \quad 6 \mapsto 2, \quad 2 \mapsto 1$$

1(c) Non-uniqueness of decomposition into transpositions

The decomposition of a cycle into transpositions is not unique. For example,

$$(1, 2, 3) = (1, 3)(1, 2).$$

Indeed, applying the right-hand side from right to left gives

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 1,$$

so it is equal to $(1, 2, 3)$.

On the other hand,

$$(1, 2, 3) = (2, 3)(1, 3).$$

Again, checking the action gives

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 1.$$

So the same cycle has at least two different decompositions into transpositions.

1(d) Relations for the adjacent transpositions

For $1 \leq i < n$, define

$$\tau_i = (i, i + 1).$$

We show that these satisfy the relations

$$\tau_i^2 = 1,$$

$$\tau_{i+1}\tau_i\tau_{i+1} = \tau_i\tau_{i+1}\tau_i,$$

$$\tau_i\tau_j = \tau_j\tau_i \quad \text{for } |i - j| \geq 2.$$

First, since τ_i swaps i and $i + 1$, applying it twice returns every element to itself. Hence

$$\tau_i^2 = 1.$$

Next, assume $|i - j| \geq 2$. Then the transpositions $\tau_i = (i, i + 1)$ and $\tau_j = (j, j + 1)$ act on disjoint pairs of elements. Therefore they commute:

$$\tau_i\tau_j = \tau_j\tau_i.$$

Finally, consider the braid relation

$$\tau_{i+1}\tau_i\tau_{i+1} = \tau_i\tau_{i+1}\tau_i.$$

Both sides fix every element except possibly i , $i + 1$, and $i + 2 \pmod n$. So it suffices to compare their action on these three elements.

For the left-hand side:

$$\begin{aligned} i &\xrightarrow{\tau_{i+1}} i \xrightarrow{\tau_i} i + 1 \xrightarrow{\tau_{i+1}} i + 2, \\ i + 1 &\xrightarrow{\tau_{i+1}} i + 2 \xrightarrow{\tau_i} i + 2 \xrightarrow{\tau_{i+1}} i + 1, \\ i + 2 &\xrightarrow{\tau_{i+1}} i + 1 \xrightarrow{\tau_i} i \xrightarrow{\tau_{i+1}} i. \end{aligned}$$

So the left-hand side sends

$$i \mapsto i + 2, \quad i + 1 \mapsto i + 1, \quad i + 2 \mapsto i.$$

For the right-hand side:

$$\begin{aligned} i &\xrightarrow{\tau_i} i+1 \xrightarrow{\tau_{i+1}} i+2 \xrightarrow{\tau_i} i+2, \\ i+1 &\xrightarrow{\tau_i} i \xrightarrow{\tau_{i+1}} i \xrightarrow{\tau_i} i+1, \\ i+2 &\xrightarrow{\tau_i} i+2 \xrightarrow{\tau_{i+1}} i+1 \xrightarrow{\tau_i} i. \end{aligned}$$

So the right-hand side has the same action. Therefore

$$\tau_{i+1}\tau_i\tau_{i+1} = \tau_i\tau_{i+1}\tau_i.$$

1(e) The sign homomorphism

Recall that a *group homomorphism* between two groups $(G, \cdot)$ and $(H, \cdot)$ is a map

$$\varphi : G \rightarrow H$$

such that

$$\varphi(gh) = \varphi(g)\varphi(h) \quad \text{for all } g, h \in G.$$

Suppose that $\sigma \in S_n$ decomposes into disjoint cycles of lengths $k_1, \dots, k_r$. Define

$$N(\sigma) = \sum_{m=1}^r (k_m - 1), \quad \text{sign}(\sigma) = (-1)^{N(\sigma)}.$$

We want to show that

$$\text{sign} : S_n \rightarrow \{\pm 1\}$$

is a group homomorphism, i.e.

$$\text{sign}(\sigma\rho) = \text{sign}(\sigma)\text{sign}(\rho).$$

Step 1: Counting transpositions. From part (b), every cycle of length k can be written as a product of exactly $k - 1$ transpositions. Therefore, if σ decomposes into disjoint cycles of lengths $k_1, \dots, k_r$, then σ can be written as a product of exactly

$$(k_1 - 1) + \dots + (k_r - 1) = N(\sigma)$$

transpositions.

So $\text{sign}(\sigma)$ records whether σ can be written as a product of an even or odd number of transpositions.

Step 2: Well-definedness. Although a permutation can be written as a product of transpositions in many different ways, the parity (even or odd number) of transpositions is independent of the chosen decomposition.

This follows from the relations in part (d):

$$\tau_i^2 = 1, \quad \tau_i\tau_j = \tau_j\tau_i \quad (|i - j| \geq 2), \quad \tau_{i+1}\tau_i\tau_{i+1} = \tau_i\tau_{i+1}\tau_i.$$

Each of these relations preserves the parity of the number of transpositions:

- $\tau_i^2 = 1$ removes or inserts two transpositions,
- the other relations do not change the number.

Hence the parity is well-defined.

Step 3: Homomorphism property. Suppose σ is a product of m transpositions and ρ is a product of n transpositions. Then $\sigma\rho$ is a product of $m+n$ transpositions.

Thus

$$\text{sign}(\sigma) = (-1)^m, \quad \text{sign}(\rho) = (-1)^n, \quad \text{sign}(\sigma\rho) = (-1)^{m+n}.$$

Therefore

$$\text{sign}(\sigma\rho) = (-1)^{m+n} = (-1)^m(-1)^n = \text{sign}(\sigma)\text{sign}(\rho).$$

Conclusion. Thus $\text{sign} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism.

Exercise 2: Chinese remainder theorem

Let a and b be coprime positive integers. We show that

$$\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z} \cong \mathbb{Z}/ab\mathbb{Z}.$$

Define a map

$$\phi : \mathbb{Z}/ab\mathbb{Z} \rightarrow \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}, \quad [x]_{ab} \mapsto ([x]_a, [x]_b).$$

This is a group homomorphism for the additive group structures, because

$$\phi([x+y]_{ab}) = ([x+y]_a, [x+y]_b) = ([x]_a, [x]_b) + ([y]_a, [y]_b).$$

We show that ϕ is injective. Suppose

$$\phi([x]_{ab}) = ([0]_a, [0]_b).$$

Then $a \mid x$ and $b \mid x$. Since a and b are coprime, it follows that $ab \mid x$. Hence

$$[x]_{ab} = [0]_{ab},$$

so ϕ is injective.

We now show that ϕ is surjective. Let $([r]_a, [s]_b)$ be any element of

$$\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}.$$

Since a and b are coprime, Bézout's identity gives integers u, v such that

$$ua + vb = 1.$$

Define

$$x = r vb + s ua.$$

Then modulo a , we have $ua \equiv 0$ and $vb \equiv 1$, so

$$x \equiv r \pmod{a}.$$

Similarly, modulo b , we have $vb \equiv 0$ and $ua \equiv 1$, so

$$x \equiv s \pmod{b}.$$

Therefore

$$\phi([x]_{ab}) = ([r]_a, [s]_b).$$

So ϕ is surjective.

Hence ϕ is an isomorphism, and therefore

$$\mathbb{Z}/ab\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}.$$

For a general positive integer n , let

$$n = p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$$

be its prime factorization. Since distinct prime powers are pairwise coprime, repeated application of the theorem gives

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{m_1}\mathbb{Z} \times \mathbb{Z}/p_2^{m_2}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_r^{m_r}\mathbb{Z}.$$

Thus any finite cyclic group decomposes as a product of its prime-power parts.

Exercise 3: Automorphisms of $\mathbb{Z}_n$

Let G be a group. We show first that the set $\text{Aut}(G)$ of invertible group homomorphisms $G \rightarrow G$ forms a group.

The operation is composition of maps. We verify the group axioms:

- If $f, g \in \text{Aut}(G)$, then $f \circ g$ is again a group homomorphism, and since both f and g are bijective, $f \circ g$ is bijective. So $f \circ g \in \text{Aut}(G)$.
- Composition of functions is associative.
- The identity map id_G is an invertible group homomorphism.
- If $f \in \text{Aut}(G)$, then f is bijective. Its inverse $f^{-1} : G \rightarrow G$ is also a group homomorphism, because for any $x, y \in G$ we can write $x = f(a)$ and $y = f(b)$, and then

$$f^{-1}(xy) = f^{-1}(f(a)f(b)) = f^{-1}(f(ab)) = ab = f^{-1}(x)f^{-1}(y).$$

So $f^{-1} \in \text{Aut}(G)$.

Therefore $\text{Aut}(G)$ is a group.

Now let us compute $\text{Aut}(\mathbb{Z}_n)$. We regard $\mathbb{Z}_n$ as an additive cyclic group generated by $[1]$. Any group homomorphism

$$f : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$$

is completely determined by the image of the generator $[1]$. If

$$f([1]) = [k],$$

then for any $m \in \mathbb{Z}$,

$$f([m]) = f(m[1]) = mf([1]) = [mk].$$

So every endomorphism of $\mathbb{Z}_n$ is “multiplication by $[k]$ ”.

Such a homomorphism is an automorphism if and only if it is bijective. We now show that this happens if and only if $\gcd(k, n) = 1$.

First, suppose that $\gcd(k, n) = 1$. Then by Bézout’s identity there exist integers u, v such that

$$uk + vn = 1.$$

Reducing modulo n , we obtain

$$uk \equiv 1 \pmod{n}.$$

Hence $[u]$ is a multiplicative inverse of $[k]$ in $\mathbb{Z}_n$.

Define a map

$$g : \mathbb{Z}_n \rightarrow \mathbb{Z}_n, \quad g([m]) = [um].$$

Then for every $[m] \in \mathbb{Z}_n$,

$$g(f([m])) = g([km]) = [ukm] = [m],$$

and similarly

$$f(g([m])) = f([um]) = [kum] = [m].$$

Thus g is the inverse of f , and hence f is an automorphism.

Conversely, suppose that $\gcd(k, n) = d > 1$. Then we can write $k = dk'$ and $n = dn'$ for some integers k', n' . Consider the element $[n'] \in \mathbb{Z}_n$. Since $0 < n' < n$, we have $[n'] \neq [0]$. However,

$$f([n']) = [kn'] = [dk'n'] = [k'n] = [0].$$

Thus f sends a nonzero element to 0, so f is not injective, and therefore not an automorphism.

We conclude that $f([m]) = [mk]$ defines an automorphism if and only if $\gcd(k, n) = 1$. Hence

$$\text{Aut}(\mathbb{Z}_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

Examples. If $n = 5$, every nonzero class is invertible, so

$$\text{Aut}(\mathbb{Z}_5) \cong (\mathbb{Z}/5\mathbb{Z})^\times = \{[1], [2], [3], [4]\},$$

a group of order 4.

If $n = 8$, the invertible classes are

$$[1], [3], [5], [7],$$

so

$$\text{Aut}(\mathbb{Z}_8) \cong (\mathbb{Z}/8\mathbb{Z})^\times,$$

which has order 4.

Exercise 4: The Lorentz group

Recall that the Lorentz group $O(3, 1)$ consists of linear transformations $\Lambda : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ preserving the bilinear form

$$-x_0^2 + x_1^2 + x_2^2 + x_3^2.$$

We are asked to analyse its decomposition into a continuous subgroup and discrete symmetries.

4(a) A homomorphism to $\mathbb{Z}_2 \times \mathbb{Z}_2$

There are two binary properties of a Lorentz transformation:

- whether it preserves time orientation or reverses it,
- whether it preserves spatial orientation or reverses it.

Define a map

$$F : O(3, 1) \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$$

by setting

$$F(\Lambda) = (a, b),$$

where

- $a = 0$ if Λ is orthochronous and $a = 1$ if it is time reversing,
- $b = 0$ if Λ is orientation preserving and $b = 1$ if it is parity reversing.

Under composition, these properties combine modulo 2: composing two time reversals gives a time-preserving map, and similarly for parity. Therefore

$$F(\Lambda\Lambda') = F(\Lambda) + F(\Lambda')$$

with addition in $\mathbb{Z}_2 \times \mathbb{Z}_2$. Hence F is a group homomorphism.

4(b) Existence of a section

We want a group homomorphism

$$s : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow O(3, 1)$$

such that

$$F \circ s = \text{id}.$$

Define

$$\begin{aligned} s(0, 0) &= I, \\ s(1, 0) &= T = \text{diag}(-1, 1, 1, 1), \\ s(0, 1) &= P = \text{diag}(1, -1, -1, -1), \\ s(1, 1) &= TP = \text{diag}(-1, -1, -1, -1). \end{aligned}$$

These matrices preserve the Lorentz form, so they belong to $O(3, 1)$. Moreover,

$$T^2 = P^2 = I, \quad TP = PT,$$

so the assignment respects the group law of $\mathbb{Z}_2 \times \mathbb{Z}_2$. Clearly,

$$F(I) = 0, \quad F(T) = (1, 0), \quad F(P) = (0, 1), \quad F(TP) = (1, 1).$$

Thus $F \circ s = \text{id}$, so s is a section.

4(c) Description of $SO(3, 1)^+$ and set decomposition

The kernel of F is called the proper orthochronous Lorentz group:

$$SO(3, 1)^+ = \ker(F).$$

By definition, its elements preserve both time orientation and spatial orientation. This is the identity component of the Lorentz group. Physically, its elements are generated by spatial rotations and Lorentz boosts.

Because s is a section, every element of $O(3, 1)$ can be written in the form

$$\Lambda s(a, b) \quad \text{with } \Lambda \in SO(3, 1)^+, (a, b) \in \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Indeed, given $g \in O(3, 1)$, let $(a, b) = F(g)$. Then

$$gs(a, b)^{-1} \in \ker(F) = SO(3, 1)^+,$$

so

$$g = \Lambda s(a, b)$$

for some $\Lambda \in SO(3, 1)^+$.

This expression is unique. If

$$\Lambda s(a, b) = \Lambda' s(a', b'),$$

then

$$\Lambda^{-1}\Lambda' = s(a, b)s(a', b')^{-1}.$$

The left-hand side lies in $SO(3, 1)^+$, while the right-hand side lies in the image of the section. Applying F shows that $(a, b) = (a', b')$, hence also $\Lambda = \Lambda'$.

Therefore, as sets,

$$O(3, 1) \cong SO(3, 1)^+ \times (\mathbb{Z}_2 \times \mathbb{Z}_2).$$

This is not necessarily an isomorphism of groups with the direct product law, because the section elements need not commute with all elements of $SO(3, 1)^+$. In general the elements $s(a, b)$ do not commute with elements of $SO(3, 1)^+$. Indeed,

$$(\Lambda s(a, b))(\Lambda' s(a', b')) = \Lambda s(a, b)\Lambda' s(a', b'),$$

and we cannot rearrange this as $\Lambda\Lambda' s(a + a', b + b')$.

4(d) The induced action on $SO(3, 1)^+$

Define

$$\varphi : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \text{Aut}(SO(3, 1)^+)$$

by

$$\varphi(a, b)(\Lambda) = s(a, b)\Lambda s(a, b)^{-1}.$$

We must show that this is a group homomorphism.

First, for fixed (a, b) , the map $\Lambda \mapsto s(a, b)\Lambda s(a, b)^{-1}$ is an automorphism of $SO(3, 1)^+$, since conjugation by an invertible element is always an automorphism, and conjugation preserves the kernel of F .

Second, for (a, b) and (a', b') we have

$$\varphi(a, b)(\varphi(a', b')(\Lambda)) = s(a, b)s(a', b')\Lambda s(a', b')^{-1}s(a, b)^{-1}.$$

Since s is a homomorphism,

$$s(a, b)s(a', b') = s(a + a', b + b').$$

Thus

$$\varphi(a, b) \circ \varphi(a', b') = \varphi(a + a', b + b').$$

So φ is a group homomorphism.

4(e) The semidirect product description

We now equip the set

$$SO(3,1)^+ \times (\mathbb{Z}_2 \times \mathbb{Z}_2)$$

with the multiplication

$$(\Lambda, (a, b)) \cdot (\Lambda', (a', b')) = (\Lambda \varphi(a, b)(\Lambda'), (a + a', b + b')).$$

We show that this group is isomorphic to $O(3,1)$.

Define

$$\Psi : SO(3,1)^+ \times (\mathbb{Z}_2 \times \mathbb{Z}_2) \rightarrow O(3,1)$$

by

$$\Psi(\Lambda, (a, b)) = \Lambda s(a, b).$$

From part (c), this map is bijective.

We now check that it is a homomorphism. Compute:

$$\begin{aligned} \Psi((\Lambda, (a, b)) \cdot (\Lambda', (a', b'))) &= \Psi(\Lambda \varphi(a, b)(\Lambda'), (a + a', b + b')) \\ &= \Lambda \varphi(a, b)(\Lambda') s(a + a', b + b'). \end{aligned}$$

Using the definition of φ , this becomes

$$\Lambda s(a, b) \Lambda' s(a, b)^{-1} s(a + a', b + b').$$

Since s is a homomorphism,

$$s(a, b)^{-1} s(a + a', b + b') = s(a', b').$$

Therefore

$$\Psi((\Lambda, (a, b)) \cdot (\Lambda', (a', b'))) = \Lambda s(a, b) \Lambda' s(a', b').$$

But this is exactly

$$\Psi(\Lambda, (a, b)) \Psi(\Lambda', (a', b')).$$

Hence Ψ is a group homomorphism.

Since Ψ is both bijective and a homomorphism, it is a group isomorphism. Therefore

$$O(3,1) \cong SO(3,1)^+ \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2).$$

A Appendix

A.1 Exercise 4(c)

We now justify that $SO(3,1)^+$ is generated by spatial rotations and Lorentz boosts.

Let $\Lambda \in SO(3,1)^+$. Consider the vector

$$v = \Lambda e_0.$$

Since Λ preserves the Minkowski form, we have

$$\langle v, v \rangle = \langle e_0, e_0 \rangle = -1,$$

so v is a timelike vector. Moreover, since Λ is orthochronous, $v_0 > 0$, so v is future-pointing.

A standard fact is that there exists a Lorentz boost B such that

$$Bv = e_0.$$

Hence

$$B\Lambda e_0 = e_0.$$

Define $\Lambda' = B\Lambda$. Then Λ' fixes e_0 , so it does not mix time with space. Therefore it must be of the form

$$\Lambda' = \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix},$$

where $R \in SO(3)$ is a spatial rotation.

Thus

$$\Lambda = B^{-1}R,$$

which shows that Λ can be written as a composition of a boost and a rotation.

Therefore $SO(3, 1)^+$ is generated by spatial rotations and Lorentz boosts.

We now justify the statement that any future timelike vector can be mapped to e_0 by a Lorentz boost (after possibly applying a spatial rotation first).

Let

$$v = (v_0, v_1, v_2, v_3) = (v_0, \vec{v})$$

be a future timelike vector. This means that

$$-v_0^2 + |\vec{v}|^2 < 0 \quad \text{and} \quad v_0 > 0.$$

Equivalently,

$$v_0^2 > |\vec{v}|^2, \quad \text{so in particular } v_0 > |\vec{v}|.$$

First apply a spatial rotation $R \in SO(3)$ sending the spatial vector $\vec{v}$ to the x_1 -axis. Extending this to a Lorentz transformation by leaving the time coordinate fixed, we obtain

$$\tilde{R}v = (v_0, r, 0, 0), \quad r = |\vec{v}|.$$

Now consider the boost in the x_1 -direction:

$$B(\eta) = \begin{pmatrix} \cosh \eta & -\sinh \eta & 0 & 0 \\ -\sinh \eta & \cosh \eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Applying this to $(v_0, r, 0, 0)$ gives

$$B(\eta) \begin{pmatrix} v_0 \\ r \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} v_0 \cosh \eta - r \sinh \eta \\ -v_0 \sinh \eta + r \cosh \eta \\ 0 \\ 0 \end{pmatrix}.$$

We choose η so that the spatial component vanishes:

$$-v_0 \sinh \eta + r \cosh \eta = 0.$$

This is equivalent to

$$\tanh \eta = \frac{r}{v_0}.$$

Since $v_0 > r$, we have $0 \leq r/v_0 < 1$, so such an η exists.

With this choice,

$$B(\eta)\tilde{R}v = (\lambda, 0, 0, 0)$$

for some $\lambda > 0$. Since Lorentz transformations preserve the Minkowski norm,

$$-\lambda^2 = -v_0^2 + r^2,$$

hence

$$\lambda = \sqrt{v_0^2 - r^2}.$$

Therefore any future timelike vector can be mapped to a vector of the form

$$(\sqrt{v_0^2 - |\vec{v}|^2}, 0, 0, 0)$$

by a composition of a spatial rotation and a boost. In particular, if

$$-v_0^2 + |\vec{v}|^2 = -1,$$

then $\lambda = 1$, and the vector is mapped exactly to

$$e_0 = (1, 0, 0, 0).$$